

Shravya Kanchi

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Ph.D. in Computer Science, GPA 3.76/4.0

Blacksburg, VA

Aug. 2021 – Present

IIIT Hyderabad

MS by Research in Computer Science, GPA 9/10.0

Hyderabad, India

Jul. 2018 – Aug. 2021

IIIT Sricity

B.Tech in Computer Science and Engineering with Honors, GPA 8.79/10.0

Sricity, India

Jul. 2014 – May. 2018

SELECTED PROJECTS

Synthetic data to strengthen security defenses

Under Submission

- Engineered a novel VQVAE-based generative AI model for numerical tabular data generation, driving a 20% performance improvement across five varied ML-based security defenses.
- Outperformed five SOTA tabular data generators—CTABGAN, TVAE, TabDDPM, REaLTabFormer, and GReaT—across diverse model families (LLMs, GANs, VAEs) on security classification tasks.
- Conducted a comprehensive measurement study, identifying three major data challenges impacting security ML defenses by analyzing 35 top-tier security research articles.

Toxicity mitigation in conversational AI training pipelines

Under Submission

- Proposed a framework for toxicity mitigation on diverse chatbot customization pipelines.
- Proposed context-aware and context-agnostic approaches leveraging LLMs to identify toxic conversations which outperformed industry API services in a data poisoning setting.
- Safety-aligned chatbots by adding synthetically crafted conversations with desirable traits through Direct Preference Optimization (DPO) achieving near-zero toxicity.

PUBLICATIONS

[ArXIV] “Lessons Learned from Integrating Generative AI into ML-based Security Tasks” 1st **author**.

[ArXIV] “A Defense Framework to Mitigate Toxicity While Fine-tuning Conversational AI” 1st **author**.

[IEEE S&P’24] “An Analysis of Recent Advances in Deepfake Image Detection in an Evolving Threat Landscape.” 3rd **author**. ([Paper Link](#))

[ACSAC’23] “First Look at Toxicity Injection Attacks on Open-domain Chatbots.” 3rd **author**. ([Paper Link](#))

[WWW Workshop’22] “SEBI Regulation Biography” 3rd **author**. ([Paper Link](#))

[CODASPY’21] “A multi perspective access control in a smart home.” 1st **author**. ([Paper Link](#))

WORK EXPERIENCE

Graduate Research Assistant

Aug. 2021 – Present

Virginia Tech

- Conducted research on secure and responsible generative AI, including synthetic tabular data generation to boost performance of security defenses, chatbot toxicity mitigation via DPO and LLM classifiers, and agent-based automation of jailbreak attacks outperforming SOTA baselines.

Graduate Research Assistant

Jan. 2021 – Jun 2021

IIIT Hyderabad & JP Morgan Chase

- Built the first named-entity labeled corpus for 7,500 SEBI sub-regulations with 7 proposed domain-specific entity types, and developed an overlapping NER tool achieving 87.47% precision.

TECHNICAL SKILLS

GenAI: LLMs, LoRA, model customization, DPO, Stable Diffusion, Transformers, Variational Autoencoder (VAE), Generative Adversarial Networks (GANs), prompt engineering, hyper-parameter search, BERT, Fine-tuning, PEFT.

Security: Malware detection, Phishing, Spam, Network IDS, concept drift, website privacy, anomaly detection.

Frameworks: Deep learning - Pytorch, Tensorflow, Keras, Flask, MongoDB, AutoML.

Developer Tools: Git, Linux, VS Code, Vim, Jupyter Notebook, Docker, Nmap, label-studio

Libraries: NumPy, Huggingface, Transformers, RayTune, Matplotlib, gnuplot, Scikit-Learn, pandas, LightGBM, SpaCy

Languages: Python, C++, C, Shell, SQL